

Table 4: Parameters estimates for stock indexes.

| Stock index | N    | $\alpha_0$ | $\alpha_1$ | $\beta$  | $\tau$    | $\theta$ | $\sigma_J$ | $q$      | ln L   | $\Omega$  |
|-------------|------|------------|------------|----------|-----------|----------|------------|----------|--------|-----------|
| CAC40       | 2798 | 0.2383     | -257.6798  | 0.1330   | 54.8009   | 0.0116   | 0.0627     | 0.1645   | 5376.7 | 344.59*** |
|             |      | ( 0.1155)  | (4.9044)   | (0.0103) | (5.0972)  | (0.0029) | (0.0043)   | (0.0285) |        |           |
|             |      | -0.0615    | -257.9421  | 0.2687   | 82.9046   | 0.0093   |            |          | 5204.4 |           |
| DAX         | 2785 | ( 0.0978)  | (6.3565)   | (0.0104) | (7.5923)  | (0.0024) |            |          |        | 280.85*** |
|             |      | 0.2784     | -255.8185  | 0.1555   | 57.6903   | 0.0127   | 0.0645     | 0.1652   | 5164.5 |           |
|             |      | ( 0.2202)  | (24.7216)  | (0.0167) | (17.3980) | (0.0031) | (0.0039)   | (0.0300) |        |           |
| DJIA        | 2703 | -0.0118    | -252.6514  | 0.2919   | 86.8931   | 0.0124   |            |          | 5024.1 | 317.71*** |
|             |      | ( 0.1394)  | (6.1700)   | (0.0110) | (9.5216)  | (0.0023) |            |          |        |           |
|             |      | 0.1067     | -210.7317  | 0.0590   | 47.0208   | 0.0082   | 0.0430     | 0.1481   | 6373.6 |           |
| FTSE100     | 2713 | ( 0.0605)  | (5.4284)   | (0.0048) | (8.5147)  | (0.0021) | (0.0035)   | (0.0263) |        | 222.60*** |
|             |      | 0.0699     | -211.6276  | 0.1050   | 66.2194   | 0.0168   |            |          | 6214.8 |           |
|             |      | ( 0.1136)  | (5.6852)   | (0.0041) | (8.1033)  | (0.0022) |            |          |        |           |
| HANGSENG    | 2702 | -0.1870    | -260.5673  | 0.0649   | 61.1198   | -0.0018  | 0.0379     | 0.3053   | 5841.7 | 366.22*** |
|             |      | (0.3935)   | (9.5712)   | (0.0387) | (31.0472) | (0.0030) | (0.0084)   | (0.2009) |        |           |
|             |      | -0.1749    | -264.5221  | 0.1641   | 79.7363   | -0.0028  |            |          | 5730.4 |           |
| NASDAQ      | 2767 | ( 0.4693)  | (18.1376)  | (0.0456) | (65.8072) | (0.0018) |            |          |        | 238.63*** |
|             |      | 0.1166     | -245.2607  | 0.1369   | 41.1688   | 0.0186   | 0.0655     | 0.2412   | 4863.4 |           |
|             |      | ( 1.5002)  | (7.6386)   | (0.1075) | (8.8111)  | (0.0167) | (0.0125)   | (0.1701) |        |           |
| SP500       | 2804 | -0.0341    | -238.4197  | 0.3550   | 81.9699   | 0.0035   |            |          | 4683.4 | 278.23*** |
|             |      | (0.1558)   | (6.8947)   | (0.0132) | (12.5799) | (0.0028) |            |          |        |           |
|             |      | 0.4242     | -253.6221  | 0.1810   | 66.7837   | 0.0047   | 0.0658     | 0.2204   | 4740.6 |           |
| SPASX200    | 2870 | (0.1197)   | (6.3600)   | (0.0143) | (9.0416)  | (0.0033) | (0.0043)   | (0.0312) |        | 389.95*** |
|             |      | -0.3926    | -255.9448  | 0.3824   | 91.8324   | 0.0070   |            |          | 4621.3 |           |
|             |      | ( 0.0814)  | (7.9904)   | (0.0177) | (19.2416) | (0.0028) |            |          |        |           |
| SMI         | 2704 | 0.1679     | -261.8900  | 0.0441   | 24.8916   | 0.0235   | 0.0360     | 0.4098   | 6122.3 | 39.95***  |
|             |      | (0.1935)   | (5.3790)   | (0.0108) | (4.8198)  | (0.0050) | (0.0022)   | (0.0706) |        |           |
|             |      | -0.0194    | -261.4696  | 0.1745   | 43.5322   | 0.0093   |            |          | 5983.2 |           |
| TWII        | 2639 | (0.1217)   | (5.5881)   | (0.0072) | (4.1993)  | (0.0031) |            |          |        | 238.07*** |
|             |      | 0.2102     | -257.0073  | 0.0570   | 71.0473   | 0.0061   | 0.0514     | 0.0785   | 6951.4 |           |
|             |      | (0.1332)   | (6.5243)   | (0.0035) | (6.9430)  | (0.0014) | (0.0052)   | (0.0180) |        |           |
| TWII        | 2639 | 0.0966     | -253.5373  | 0.0908   | 112.8947  | 0.0093   |            |          | 6756.5 | 39.95***  |
|             |      | (0.0567)   | (6.6217)   | (0.0034) | (8.9553)  | (0.0011) |            |          |        |           |
|             |      | 0.1878     | -255.8562  | 0.0000   | 0.1737    | 0.8342   | 0.0526     | 0.1877   | 5684.6 |           |
| TWII        | 2639 | (0.1344)   | (4.5107)   | (0.0386) | (0.0315)  | (0.1503) | (0.0037)   | (0.0266) |        | 238.07*** |
|             |      | 0.0394     | -247.5534  | 0.1492   | 113.4965  | 0.0079   |            |          | 5664.6 |           |
|             |      | (0.1479)   | (6.3341)   | (0.0062) | (12.8102) | (0.0014) |            |          |        |           |
| TWII        | 2639 | -0.0621    | -240.4190  | 0.1094   | 21.5185   | 0.0008   | 0.0508     | 0.4276   | 4836.2 | 238.07*** |
|             |      | (0.1073)   | (5.0817)   | (0.0153) | (3.9170)  | (0.0065) | (0.0023)   | (0.0457) |        |           |
|             |      | -0.0273    | -245.3496  | 0.3676   | 31.7291   | 0.0037   |            |          | 4717.1 |           |
| TWII        | 2639 | (0.4595)   | (5.6751)   | (0.0111) | (2.9732)  | (0.0056) |            |          |        | 238.07*** |

\*\*\* Indicates significance at 1% level.

<sup>a</sup> Standard errors of maximum likelihood estimates are in parentheses.